A Finite-Element Variable Time-Stepping Algorithm for Solving the Electromagnetic Diffusion Equation

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This paper presents a new methodology for applying the Backward Differentiation Formula (BDF) and the Theta algorithm for the variable time-stepping finite-element discretization. The developed BDF-Theta formulation was implemented for solving the Electromagnetic Diffusion equation. An initial-guess prediction algorithm was adopted for convergence acceleration. An algorithm for the selection of the order and time-step size is also included. A minimum time step criterion is adopted. The proposed methodology was validated against an analytical solution and good results were obtained. The adaptive time-step (BDF) requires less computation time than using the fixed time-step implicit Euler.

***Index Terms*—Backward differentiation formula, finite elements, numerical methods, time-stepping.**

I. Introduction

The finite element method (FEM) has proved to be one of the most effective numerical solution techniques for space and time domain PDE's. Numerical methods based on forward and backward Euler and Runge-Kutta methodologies stand as the widely used ones for solving the Electromagnetic Diffusion Equation (EDE)[1]–[6]. The Backward Differentiation Formula (BDF) is a variable time-stepping method which is A-stable and L-stable and it is recommended for solving stiff problems [7], [8]. The BDF methodology has been treated as an implicit Euler problem for solving EDE's [9]. The Theta algorithm is a non-variable time-stepping method, where the parameter θ is chosen so that $0 \leq \theta \leq 1$ and different solution schemes are obtained [10], [11].

In this paper, a BDF-Theta strategy is proposed and adapted to the FEM. The paper presents a formulation that combines the damping characteristics of the theta algorithm with the variable time-step and variable order of the BDF algorithm. The methodology is applied to the transient solution of the diffusion equation using 2-D first-order triangular finite elements (FE) and it is validated against an analytical solution. The resulting nonlinear BDF equations are solved using the Newton-Raphson (NR) method where an initial prediction method is included for convergence acceleration. An algorithm is presented to select the backward differentiation order and time-step value. A suitable FE minimal time-step criterion is adopted to avoid the presence of small time steps and instability [8], [11]. The developed algorithm was programmed in C language.

II. Finite Element Model

The electromagnetic diffusion equation is given by (1)

$$\frac{1}{\mu}\nabla^2 A = -J + \sigma\frac{\partial A}{\partial t} \tag{1}$$

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where A stands for magnetic vector potential, J is the current density, μ is the magnetic permeability, t the time and σ represents the electric conductivity.

Equation (2) shows the spatial FE discretization of (1)

$$[T_{ij}]\left\{\frac{\partial A_j}{\partial t}\right\} = [M_{ij}]\{A_j\} + \{f_i\} \tag{2}$$

where

$$[T_{ij}] = \int_\Omega \sigma N_i N_j d\Omega \tag{3}$$

$$[M_{ij}] = -\int_\Omega \frac{1}{\mu}\left(\frac{\partial N_i}{\partial x}\frac{\partial N_j}{\partial x} + \frac{\partial N_i}{\partial y}\frac{\partial N_j}{\partial y}\right) d\Omega \tag{4}$$

$$\{f_i\} = \int_\Omega J N_i d\Omega \tag{5}$$

where i *and* j *stand for node numbers* and $N_{i,j}$ represent the 2-D first-order triangular FE shape functions.

By introducing a parameter θ in (2) such that

$$A_j^{n+\theta} = \theta A_j^{n+1} + (1-\theta)A_j^n \tag{6}$$

$$f_j^{n+\theta} = \theta f_j^{n+1} + (1-\theta)f_j^n \tag{7}$$

The Theta scheme (8) is obtained as follows:

$$[T_{ij}]\left\{\frac{\partial A_j^{n+\theta}}{\partial t}\right\} = [M_{ij}]\left\{A_j^{n+\theta}\right\} + \left\{f_i^{n+\theta}\right\}. \tag{8}$$

By using a Taylor series, a numerical approximation of the temporal partial differentiation in (8) is obtained

$$\frac{\partial A_j^{n+\theta}}{\partial t} = \frac{A_j^{n+1} - A_j^n}{\Delta t} + O(\Delta t) \tag{9}$$

where Δt is the time step, n is time-step number and $O(\Delta t)$ represents higher order terms which are neglected.

The resulting equation from (8) and (9) gives accurate and stable results for $\theta > 1/2$ and becomes unstable for $\theta \leq 1/2$ and $\Delta t \to 0$ [10]. The theta algorithm involves fixed time steps leading to large computation times [3]–[5]. On the other hand, the BDF algorithm is a general variable time-step numerical approach that can lead to savings in computational cost. The BDF has good stability properties for orders less than sixth [7]. The implementation of backward differentiation formulas leads to a set of nonlinear algebraic equations [7], [8]. The set of nonlinear equations are solved by the NR. The NR method requires the residual and Jacobian calculation. The discrete residual and Jacobian obtained from (8) are expressed by (10) and (11) respectively

$$\{R_i\} = [M_{ij}]\left\{A_j^{n+\theta}\right\} + \left\{f_i^{n+\theta}\right\} - [T_{ij}]\left\{\frac{\partial A_j}{\partial t}\right\} \tag{10}$$

$$[Jac_{i\beta}] = \theta[M_{i\beta}] - \frac{\partial}{\partial A_\beta^{n+1}}[T_{ij}]\left\{\frac{\partial A_j}{\partial t}\right\} \tag{11}$$

where $\beta = 0, 1 \ldots$ denotes FE nodes.

The system given by (10) and (11) is solved for the variables ΔA_β^{n+1}. In the next section, a BFD numerical approximation to the temporal partial differentiation of (10) and (11), that differs from that shown in (9), is presented. For this reason, the index $n+\theta$ in (10) and (11) has been removed from the temporal partial differentiation.

III. Backward Differentiation Formulae in the FEM

The BDF implementation in the FEM presents certain advantages: 1) The arithmetic operations are smaller in comparison with the Runge-Kutta methodologies[1]–[6], the coefficient constants in the backward polynomials are common to all variables A_j in the FEM, hence, they are only calculated once. 2) The error control and prediction quantities are flexible [8]. The adaptive time-step size methods, as the implicit backward Euler, use complex and multiple error criterions for numerical stability [6]. While in the BDF, the prediction quantities are useful in the local truncation error, order selection and time-step size calculation that reduce the computational cost. Besides, the BDF does not require an evolving mesh strategy as it is needed in some space-time adaptative methods [12]. 3) Numerical experiments demonstrate that the BDF becomes unstable for fast changes in Δt [8]. However, the BDF is more stable than other methods as the Gear-Nordsieck [8] and Crank Nicholson [2].

A. Backward-Differentiation Polynomial Approximation

The partial differentiation of (10) and (11) requires the unknowns A to be time functions instead of (9). To achieve this, let A be a polynomial of k degree as

$$A(t) = \sum_{g=0}^{k} A^{n+1-g} p_{n+1-g}(t) \tag{12}$$

where p refers to time polynomials.

The order k represents how many history A values are required to evaluate (12) at a specific time. However, the temporal differentiation in (10) and (11) is evaluated at t_{n+1} [7], [8], i.e.,

$$\frac{\partial}{\partial t}A^{n+1} = -\frac{1}{\Delta t}\sum_{g=0}^{k} A^{n+1-g}\alpha_g(t_{n+1}) \tag{13}$$

where $\Delta t = t_{n+1} - t_n$ and α_g are coefficients dependent on time t_{n+1-k} and they are defined as

$$\alpha_g(t_{n+1}) = -\Delta t \frac{\partial p_{n+1-g}(t_{n+1})}{\partial t}. \tag{14}$$

The α_g values are obtained by choosing the following polynomials:

$$A(t) = \frac{1}{\Delta t^g}[t_{n+1} - t]^g. \tag{15}$$

The algebraic process to obtain α_g from (13) and (15) is described in [8]. Once (13) and (15) are solved the α_g values are calculated by using (16) and (17)

$$\alpha_g = \frac{(t_{n+1} - t_n)}{(t_{n+1} - t_{n+1-g})} \prod_{\substack{\psi=1 \\ \psi \neq g}}^{k} \frac{(t_{n+1} - t_{n+1-\psi})}{(t_{n+1-g} - t_{n+1-\psi})} \tag{16}$$

for $0 < g \leq k$, while for $g = 0$

$$\alpha_0 = -\sum_{\psi=1}^{k} \alpha_\psi. \tag{17}$$

Equations (16) and (17) store history information at t_{n+1-k} which is used to evaluate the partial differentiation of (10) and (11) at each time step. The residual (10) and Jacobian (11) expressed as function of backward polynomials in (13) are given by (18) and (19) respectively

$$\{R_i\} = [M_{ij}]\left\{A_j^{n+\theta}\right\} + \left\{f_i^{n+\theta}\right\} + [T_{ij}]\left\{\frac{1}{\Delta t}\sum_{g=0}^{k} A_j^{n+1-g}\alpha_g\right\} \tag{18}$$

$$[Jac_{i\beta}] = \theta[M_{i\beta}] + [T_{i\beta}]\left\{\frac{\alpha_0}{\Delta t}\right\}. \tag{19}$$

The solution of the above system is obtained by applying (20)

$$[Jac_{i\beta}]\left\{\Delta A_\beta^{n+1}\right\} = -\{R_i\}. \tag{20}$$

From (20) distinct schemes of solutions are obtained depending on θ, e.g., the implicit scheme of the proposed methodology leads to the common BDF-Implicit solver strategies, as the ones used in the algorithm DASSL [7].

B. Initial Guess Prediction Ap^{n+1} for the A^{n+1} Potential

The convergence in (20) accelerates when an initial guess of the unknown variable A^{n+1} is chosen to be near to the solution of the set of equations. Therefore, an advantage is taken by using the past values of A to predict the initial guess [8]. According to [8], a k order initial guess approximation of A^{n+1} could be defined as

$$Ap^{n+1} = \sum_{g=0}^{k} A^{n-g} p_{n-g}(t_{n+1}). \tag{21}$$

Similarly from [8], an alternative representation of (21) is given by

$$Ap^{n+1} = \sum_{g=0}^{k} A^{n-g}\gamma_{g+1} \tag{22}$$

where

$$\gamma_{g+1} = p_{n-g}(t_{n+1}). \tag{23}$$

Therefore, the linearization of (15) is necessary to obtain the coefficients γ_{g+1}

$$\gamma_{g+1} = \prod_{\substack{\psi=1 \\ \psi \neq (g+1)}}^{k+1} \frac{(t_{n+1} - t_{n+1-\psi})}{(t_{n-g} - t_{n+1-\psi})}. \tag{24}$$

The constants α_g and γ_{g+1} are common to all variables A_j. Therefore, they are only calculated once unless the order k changes.

IV. Selection of the k Order and Time-Step Size

A proper selection of a local truncation error E and order k determines the stability of the BDF algorithm [7], [8]. A satisfactory local truncation error is shown in [8], which is given by

$$E_k^j = \frac{\Delta t}{t_{n+1} - t_{n-k}}\left(A_j^{n+1} - Ap_j^{n+1}\right). \tag{25}$$

The derivation of the local truncation error (25) is shown in [8]. However in (25), a total absolute error E_k^j allowed on the jth unknown must be specified. There is no optimum error criterion for all problems but a satisfactory one and suitable for FE is given in [8]. The Gear's algorithm has a suitable technique for changing Δt and k based on (25). The Gear's algorithm changes the order k up or down by defining a time-step ratio η_k (26)

$$\eta_k = \frac{\Delta t_{new}}{\Delta t} = \frac{t_{n+2} - t_{n+1}}{t_{n+1} - t_n}. \tag{26}$$

The time-step ratio in (26) is obtained from (27)

$$E_k^j \left(\eta_k^j\right)^{k+1} = \frac{\Delta A_j \eta_k^j \Delta t}{t_{n+1} - t_{n-k}} \tag{27}$$

where ΔA_j is an absolute error (error control) allowed in the jth unknown variable.

The maximum time-step ratio per order k is chosen as $\eta_k = \min_{j \in \mathrm{NOD}} \eta_k^j$ where NOD refers to the set of nodes in the FE mesh. The new time-step Δt_{new} is obtained once η_k is calculated from (27). The order k is obtained by comparing the quantities η_k, η_{k-1} and η_{k+1} and selecting $\max_{k-1 \le Y \le k+1} \eta_Y$. To change the order, a new set of predicted values at the $k-1$ and $k+1$ order are necessary and they are obtained from the predicted constants in (22) as

$$\begin{aligned} \gamma_d^{k+1} &= \left(\frac{t_{n+1} - t_{n-k-1}}{t_{n+1-d} - t_{n-k-1}}\right) \gamma_d^k \quad d = 1, 2 \ldots, k+1 \\ \gamma_{k+2}^{k+1} &= 1 - \sum_{\nu=1}^{k+1} \gamma_\nu^{k+1} \end{aligned} \tag{28a}$$

for increasing order, while for decreasing order we get

$$\gamma_d^{k-1} = \left(\frac{t_{n+1-d} - t_{n-k}}{t_{n+1} - t_{n-k}}\right) \gamma_d^{k-1} \quad d = 1, 2 \ldots, k. \tag{28b}$$

Similarly, the decreasing and increasing formulas are used to define the coefficients α_g for different orders. For an increasing order, we get

$$\begin{aligned} \alpha_d^{k+1} &= \frac{(t_{n+1} - t_{n-k})}{(t_{n+1-d} - t_{n-k})} \alpha_d^k \quad d = 1, 2 \ldots, k \\ \alpha_0^{k+1} &= \alpha_0^k - \frac{(t_{n+1} - t_n)}{(t_{n+1} - t_{n-k})} \\ \alpha_{k+1}^{k+1} &= -\sum_{\nu=0}^{k} \alpha_\nu^{k+1} \end{aligned} \tag{29a}$$

while for a decreasing order

$$\begin{aligned} \alpha_q^{k-1} &= \frac{(t_{n+1-q} - t_{n+1-k})}{(t_{n+1} - t_{n+1-k}} \alpha_q^k \quad q = 1, 2 \ldots, k-1 \\ \alpha_0^{k-1} &= \alpha_0^k + \frac{(t_{n+1} - t_n)}{(t_{n+1} - t_{n+1-k})}. \end{aligned} \tag{29b}$$

The use of a minimum time-step constraint avoids the presence of interruptions in the smoothness of the results and brings numerical stability [8]. This paper proposes the minimal time step strategy given by (30)

$$\Delta t_{\min} = C \mu \sigma h^2 \tag{30}$$

where C is a constant chosen so that $C < 1$ and h is related to the minimal FE size [11].

V. Simulation Results

The BDF-Theta algorithm, as given by (18) and (19) was programmed in C language and solved for the domain shown in Fig. 1. The FE mesh has 12022 nodes and 23786 first order elements. The model was executed on an HP computer, with 2 GHz dual core and 2 GB RAM. To demonstrate the validity of the proposed formulation, two numerical experiments were made. The numerical solution of each experiment was compared against an analytical solution and a fixed time-step method (implicit Euler method). The analytical eddy-current distribution (Fig. 1) for $y \le 0$ is given by (31) [13]

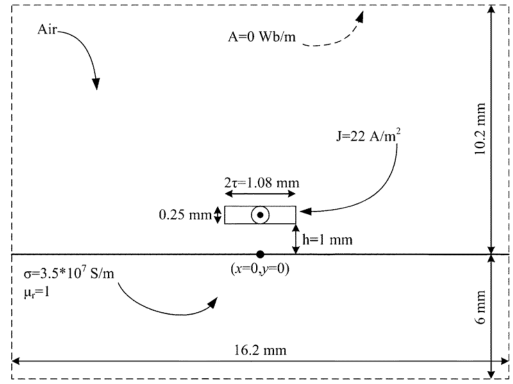

Fig. 1. Domain used in the FE time-stepping algorithm.

$$\begin{aligned} J(x,y,t) = \frac{-I_0}{\pi} \int_0^\infty & \left\{ \frac{\sin(\delta\tau)\cos(\delta x)}{\delta\tau} e^{-\delta h} \exp\left(\frac{-\delta^2 t}{\sigma\mu_0\mu_r}\right) \right\} \\ & \cdot \left\{ \left(\sqrt{\frac{\sigma\mu_0\mu_r}{\pi t}} \exp\left(\frac{-y^2 \sigma\mu_0\mu_r}{4t}\right) \right) - (\delta \exp(-\delta\mu_r y) \right. \\ & \left. \cdot \exp\left(\frac{\delta^2 \mu_r t}{\sigma\mu_0}\right) erfc\left(\frac{-y}{2}\sqrt{\frac{\sigma\mu_0\mu_r}{t}} + \delta\mu_r \sqrt{\frac{t}{\sigma\mu_0\mu_r}}\right)\right) \right\} d\delta. \end{aligned} \tag{31}$$

The transient solution by using the implicit Euler method required 1571 s of computation time with constant time step of 1E-7 s.

A. Experiment 1: Transient Solution for Error Control Values From 1E-2 to 1E-7 and a Δt_{min} is Neglected

The transient eddy-current density at different penetration depths for $\theta = 1$ predicted by the numerical and analytical models is shown in Fig. 2. It was found that for errors from 1E-2 to 1E-3 the transient solution was better for a tighter error control. For an error range from 1E-4 to 1E-7, the transient solution was worst. This experiment demonstrates that for tighter error criterions, the transient solution is worst than that under less restrictive conditions when $\theta = 1$ [8]. It was observed that a sudden variation in Δt caused the above numerical results [8]. As it is explained in [8], it would seem that there are certain restrictions of Δt which would be sufficient for having stability and convergence. It was observed that decreasing theta from $\theta = 1$ to $\theta = 0.75$ appears to be sufficient for good numerical accuracy. However, the simulation results showed that the BDF-Theta methodology becomes unstable when $\theta = 0.75$ and with an error of 1E-7.

B. Experiment 2: Transient Solution for Error Control Values of 1E-2 to 1E-7 Considering a Δt_{min} With $C = 0.5$

It was observed that the Δt_{min} constraint used in the BDF-Theta algorithm filters out the fast transients that causes instability and divergence with respect to the Euler method. However, the computation time is larger for tighter error values. The results obtained with the BDF-Theta method $(\theta = 1)$ are close to those achieved by using the implicit Euler method when the

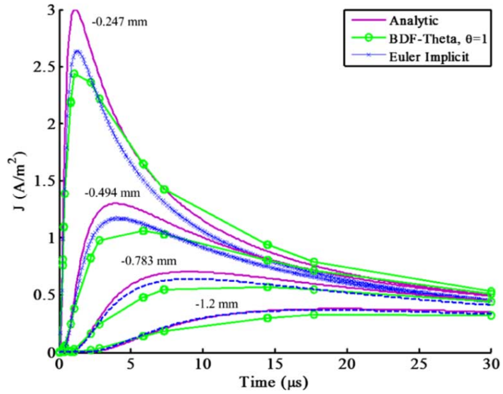

Fig. 2. Eddy current density in the transient solution for $\theta = 1$ at different penetration depths where a minimum time-step constraint is neglected.

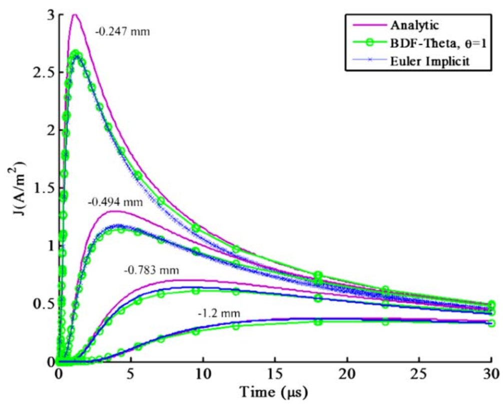

Fig. 3. Eddy current density in the transient solution for $\theta = 1$ at different penetration depths where a minimum time-step constraint is adopted.

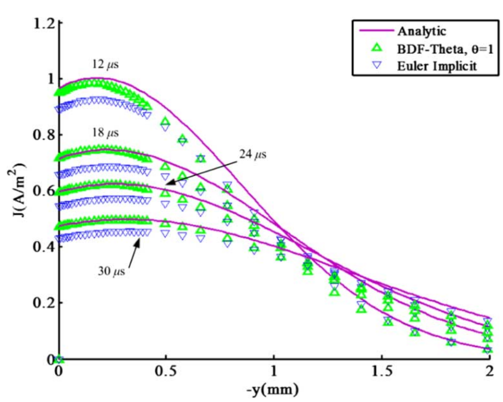

Fig. 4. Spatial eddy-current distribution for $\theta = 1$ at different times and taking into account the minimum time-step constraint.

Δt_{min} constraint is employed (Fig. 3). The spatial eddy-current distributions along the negative y-axis of the problem (Fig. 1) at a specific time are shown in Fig. 4, where it is clearly seen that the proposed methodology gives closer results to the analytical solution than the implicit Euler does.

Finally, the maximum order k and the computation time in the experiments, at each error value and with $\theta = 1$, are given in Table I. The implicit Euler took 1571 s while the maximum BDF time took 454 s, which means that BDF is 3.4 times faster than Euler. The numerical experiments showed that the BDF-Theta algorithm is more stable when a Δt_{min} restriction is taken. For an error of 1E-7, $\theta = 0.75$ and Δt_{min} restriction the BDF-Theta algorithm did not present an instability as it occurs in experiment 1 for $\theta = 0.75$ and 1E-7.

TABLE I
COMPUTATION TIME AND MAXIMUM ORDER OBTAINED IN THE NUMERICAL EXPERIMENTS FOR $\Theta = 1$

Error control	1E-4		1E-5		1E-6		1E-7	
Experiment	E1	E2	E1	E2	E1	E2	E1	E2
Order (K_{max})	2	2	2	2	2	3	2	4
Computation time (s)	89	124	131	146	136	213	137	454

VI. CONCLUSION

A FE model with variable time-step and order, which is based on the BDF and Theta formulation, has been proposed. The BDF methodology allows speeding up computation time when compared to the fixed time-step implicit Euler method. The algorithm was compared against an analytical solution and good agreement was obtained.

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